

Creating Complex Tables

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1 Brief Introduction of NHANES dataset

The National Health and Nutrition Examination Survey (NHANES) collects comprehensive health and nutrition information from the U.S. population (National Center for Health Statistics, n.d.). This report provides a descriptive summary of the cleaned NHANES dataset used in this project.

The tables summarize the sample across NHANES waves, including sample size, sex distribution, and ethnicity. Summary statistics for average systolic blood pressure (SBP) are also presented for each survey wave.

2 Load Packages and Import Dataset

```
library(tidyverse)
library(rio)
library(here)
library(knitr)
library(kableExtra)
```

```
nhanes_data <- import(
  here("data", "cleaned_NHANES.csv")
)
```

3 Overall Sample Size by Wave

The table below presents the total number of participants included in each NHANES wave, providing an overview of the sample size across survey periods.

```
sample_size_by_wave <- nhanes_data %>%
  group_by(wave_id) %>%
  summarise(
    sample_size = n(),
    .groups = "drop"
  )

kable(
  sample_size_by_wave,
  format = "latex",
  booktabs = TRUE,
  caption = "Overall Sample Size by NHANES Wave",
  col.names = c("NHANES Wave", "Sample Size")
) %>%
  kable_styling(
    latex_options = "hold_position"
  ) %>%
  column_spec(
    1,
    bold = TRUE,
    width = "3cm"
  ) %>%
  column_spec(
    2,
    width = "3cm"
  ) %>%
  footnote(
    general = "Sample size represents the number of participants included in each NHANES wave."
  )
```

Table 1: Overall Sample Size by NHANES Wave

NHANES Wave	Sample Size
2015	10175
2016	9971
2017	9254

Note:

Sample size represents the number of participants included in each NHANES wave.

4 Number and Percentage of Male Participants by Wave

The table below presents the number and percentage of male participants across NHANES waves. Percentages are calculated within each wave to provide a consistent basis for comparing the male participant distribution across survey periods.

```
male_by_wave <- nhanes_data %>%
  group_by(wave_id) %>%
  summarise(
    sample_size = n(),
    male_n = sum(gender == "Male", na.rm = TRUE),
    male_percent = 100 * male_n / sample_size,
    .groups = "drop"
  ) %>%
  select(wave_id, male_n, male_percent)

kable(
  male_by_wave,
  format = "latex",
  booktabs = TRUE,
  digits = 1,
  caption = "Number and Percentage of Male Participants by NHANES Wave",
  col.names = c("NHANES Wave", "Male n", "Male (%)")
) %>%
  add_header_above(
    c(" " = 1, "Male Participants" = 2)
  ) %>%
  kable_styling(
    latex_options = "hold_position"
  ) %>%
  column_spec(
    1,
    bold = TRUE,
    width = "3cm"
  ) %>%
  column_spec(
    2:3,
    width = "3cm"
  ) %>%
  row_spec(
    0,
    bold = TRUE
  ) %>%
  footnote(
    general = "Percentages are calculated within each NHANES wave."
  )
)
```

5 Number and Percentage of Participants by Ethnicity and Wave

The table below presents the number and percentage of participants in each ethnicity category across NHANES waves. Percentages are calculated within each wave to facilitate comparisons of the ethnicity distribution across survey periods.

Table 2: Number and Percentage of Male Participants by NHANES Wave

NHANES Wave	Male Participants	
	Male n	Male (%)
2015	5003	49.2
2016	4892	49.1
2017	4557	49.2

Note:

Percentages are calculated within each NHANES wave.

```

ethnicity_by_wave <- nhanes_data %>%
  filter(!is.na(ethnicity_2)) %>%
  group_by(wave_id, ethnicity_2) %>%
  summarise(
    n = n(),
    .groups = "drop"
  ) %>%
  group_by(wave_id) %>%
  mutate(
    percent = 100 * n / sum(n)
  ) %>%
  ungroup()

kable(
  ethnicity_by_wave,
  format = "latex",
  booktabs = TRUE,
  digits = 1,
  caption = "Ethnicity Distribution by NHANES Wave",
  col.names = c(
    "NHANES Wave",
    "Ethnicity",
    "n",
    "Percent"
  )
) %>%
add_header_above(
  c(
    " " = 1,
    "Ethnicity" = 1,
    "Participant Distribution" = 2
  )
) %>%
kable_styling(
  latex_options = "hold_position"
) %>%
column_spec(
  1,
  bold = TRUE,
  width = "3cm"
) %>%

```

```

column_spec(
  2,
  width = "4cm"
) %>%
column_spec(
  3:4,
  width = "2.5cm"
) %>%
row_spec(
  0,
  bold = TRUE
) %>%
footnote(
  general = "Percentages are calculated within each NHANES wave."
)

```

Table 3: Ethnicity Distribution by NHANES Wave

NHANES Wave	Ethnicity	Participant Distribution	
		n	Percent
2015	Asian	1074	10.6
2015	Black	2267	22.3
2015	Mexican	1730	17.0
2015	Other	470	4.6
2015	Other Hispanic	960	9.4
2015	White	3674	36.1
2016	Asian	1042	10.5
2016	Black	2129	21.4
2016	Mexican	1921	19.3
2016	Other	505	5.1
2016	Other Hispanic	1308	13.1
2016	White	3066	30.7
2017	Asian	1168	12.6
2017	Black	2115	22.9
2017	Mexican	1367	14.8
2017	Other	634	6.9
2017	Other Hispanic	820	8.9
2017	White	3150	34.0

Note:

Percentages are calculated within each NHANES wave.

6 Summary Statistics of Average Systolic Blood Pressure by Wave

Average systolic blood pressure (SBP) was calculated using the available systolic blood pressure measurements for each participant. Missing measurements were excluded from the calculation. Participants with no available measurements were assigned a missing value.

```

nhanes_data <- nhanes_data %>%
  mutate(
    average_sbp = rowMeans(
      select(
        .,
        systolic_bp_1,
        systolic_bp_2,
        systolic_bp_3,
        systolic_bp_4
      ),
      na.rm = TRUE
    ),
    average_sbp = if_else(
      is.nan(average_sbp),
      NA_real_,
      average_sbp
    )
  )

```

The following table summarizes the minimum, maximum, median, mean, and standard deviation of average systolic blood pressure within each NHANES wave.

```

sbp_by_wave <- nhanes_data %>%
  group_by(wave_id) %>%
  summarise(
    minimum = if (all(is.na(average_sbp))) NA_real_
      else min(average_sbp, na.rm = TRUE),
    maximum = if (all(is.na(average_sbp))) NA_real_
      else max(average_sbp, na.rm = TRUE),
    median = median(average_sbp, na.rm = TRUE),
    mean = mean(average_sbp, na.rm = TRUE),
    standard_deviation = sd(average_sbp, na.rm = TRUE),
    .groups = "drop"
  )

kable(
  sbp_by_wave,
  format = "latex",
  booktabs = TRUE,
  digits = 1,
  caption = "Summary Statistics of Average Systolic Blood Pressure by NHANES Wave",
  col.names = c(
    "NHANES Wave",
    "Minimum",
    "Maximum",
    "Median",
    "Mean",
    "Standard Deviation"
  )
) %>%
  add_header_above(
    c(
      " " = 1,

```

```

    "Average Systolic Blood Pressure (mm Hg)" = 5
  )
) %>%
kable_styling(
  latex_options = "hold_position"
) %>%
column_spec(
  1,
  bold = TRUE,
  width = "3cm"
) %>%
column_spec(
  2:6,
  width = "2.5cm"
) %>%
row_spec(
  0,
  bold = TRUE
) %>%
footnote(
  general = "Average systolic blood pressure was calculated from available systolic blood pressure measurements."
)

```

Table 4: Summary Statistics of Average Systolic Blood Pressure by NHANES Wave

NHANES Wave	Average Systolic Blood Pressure (mm Hg)				
	Minimum	Maximum	Median	Mean	Standard Deviation
2015	64.7	228.7	115.3	118.3	18.1
2016	74.0	231.3	117.3	120.4	18.4
2017	72.7	234.0	118.0	121.7	20.3

Note:

Average systolic blood pressure was calculated from available systolic blood pressure measurements. Missing measurements were excluded.

7 Summary of Findings

The main findings of the analysis are summarized below:

- **Sample Size:** The NHANES sample includes participants from multiple survey waves, with sample sizes varying across waves.
- **Sex Distribution:** The proportion of male participants varies across survey waves. The table presents both the number and percentage of male participants within each wave.
- **Ethnicity Distribution:** The ethnicity table shows the number and percentage of participants in each cleaned ethnicity category across NHANES waves, allowing comparisons while accounting for differences in sample size.
- **Systolic Blood Pressure:** Average systolic blood pressure varies across NHANES waves. The minimum, maximum, median, mean, and standard deviation describe the distribution and variability of SBP in each wave.

- **Interpretation:** These results provide a descriptive overview of the NHANES sample and do not, by themselves, establish causal relationships or statistically significant differences between survey waves.

8 References

References are generated from the BibTeX entries in `citations/references.bib` cited throughout the report. The bibliography file should contain an entry with the citation key `nhanes`.

National Center for Health Statistics. n.d. *National Health and Nutrition Examination Survey*. Centers for Disease Control; Prevention. <https://www.cdc.gov/nchs/nhanes/>.